A FINITELY PRESENTED MAP OF AFFINE SCHEMES TAKING A LOCALLY CLOSED SUBSET TO A SET THAT IS NOT A FINITE UNION OF LOCALLY CLOSED SUBSETS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The counted proof adds no mathematics of its own; the shift-self-identification of B with B cap (y_1 = 0) is the pivot of the context induction.

Editorial note. The label lemma-no-chevalley names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 2112-2164: over a field k of characteristic 0 the projection f: X = Spec(k[t, x_1, x_2, ..., y_1, y_2, ...]) -> Spec(k[x_1, x_2, ..., y_1, y_2, ...]) = Y is of finite presentation; Z is the closed subset cut out by x_1 (t - 1) = 0, x_2 (t - 1)(t - 2) = 0, x_3 (t - 1)(t - 2)(t - 3) = 0, ...; U = union of the opens U_j where y_j (t - 1)...(t - j) is nonzero; on each U_j the image f(Z cap U_j) is the set of points where x_1, ..., x_j are zero and y_j is nonzero
- Context claim and proof: B = f(Z cap U) = union_j f(Z cap U_j) is not a finite union of locally closed subsets of Y; the induction on n shows any cover B = A_1 union ... union A_m has m >= n for all n (the closure of B is (x_1 = 0), some A_i is open in (x_1 = 0) but cannot contain a point with y_1 = 0 since for points of B y_1 = 0 forces x_2 = 0; the right shift map x_i -> x_{i+1}, y_i -> y_{i+1} identifies B with B cap (y_1 = 0))
- Chevalley's theorem (Algebra, theorem-chevalley; Morphisms, section-constructible) gives constructibility of images of constructible sets under finitely presented morphisms - the positive backdrop stated in the section preamble

Source reference. Original theorem 00FE: sources/stacks-chow/tags.txt, source lines 568–568. Prerequisite pointer invoked at 2179; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 054H: sources/stacks-chow/tags.txt, source lines 6640–6640. Prerequisite pointer invoked at 2180; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Chevalley's theorem says that the image of a constructible set by a finitely presented morphism of affine schemes is constructible, see Algebra, Theorem 00FE

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and Morphisms, Section 054H. We will see the same thing does not hold for images of locally closed subsets.

Let k be a field of characteristic 0. Consider the projection morphism

$$f : X = \mathrm{Spec}(k[t, x_1, x_2, \ldots, y_1, y_2, \ldots]) \longrightarrow \mathrm{Spec}(k[x_1, x_2, \ldots, y_1, y_2, \ldots]) = Y$$

This is a morphism of finite presentation. Let Z be the closed subset of X defined by

$$x_1(t-1) = 0, \quad x_2(t-1)(t-2) = 0, \quad x_3(t-1)(t-2)(t-3) = 0, \quad \ldots$$

Let $U = \bigcup_{j \geq 1} U_j$ be the open of X defined by

$$U_j = \text{points where } y_j(t-1)(t-2)...(t-j) \text{ is nonzero}$$

Then we have

$$f(Z \cap U_j) = \text{points where } x_1, \ldots, x_j \text{ are zero and } y_j \text{ is nonzero}$$

We claim that $B = f(Z \cap U) = \bigcup_{j \geq 1} f(Z \cap U_j)$ is not a finite union of locally closed subsets of Y.

Proof of the claim. Suppose that $B = A_1 \cup \cdots \cup A_m$ is a finite cover of B by locally closed subsets of Y. We will show by induction on n that $m \geq n$. The base case $n = 1$ is OK as B is nonempty. Assume $n > 1$ and that the induction hypothesis holds for $n - 1$. Since the closure of B is $(x_1 = 0)$, one of the A_i must contain some nonempty open subset of $(x_1 = 0)$. Then A_i must be open in $(x_1 = 0)$. But any such open subset cannot contain a point with $y_1 = 0$; indeed, for points of B, $y_1 = 0$ forces $x_2 = 0$, and this shows B contains no neighborhood of (x, y) inside $(x_1 = 0)$. Therefore, the remaining $m - 1$ elements restrict to a constructible cover of $B \cap (y_1 = 0)$. However, observe that the right shift map $x_i \mapsto x_{i+1}$, $y_i \mapsto y_{i+1}$ identifies B with $B \cap (y_1 = 0)$! Thus by induction hypothesis, we see that $m - 1 \geq n - 1$ and we conclde $m \geq n$. This finishes the proof of the induction step and thereby establishes the claim.

Lemma 0.1. *There exists a morphism $f : X \to Y$ of finite presentation between affine schemes and a locally closed subset T of X such that $f(T)$ is not a finite union of locally closed subsets of Y.*

Proof. See discussion above. □